

Wealthy Minds - Financial Behavior Analysis and Prediction Tool

Higher National Diploma in Software Engineering



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Selected Data Structure: Graph Implementation



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Introduction

Wealthy Minds is an intelligent personal finance application designed to help students and young adults properly understand and manage their finances. It allows users to enter their income, expenses, savings, loans, subscriptions, financial goals which is then converted to financial statements, a financial health score, spending behavior insights, and personalized predictions. Unlike traditional solutions, Wealthy minds users understand their spendings, identify potential financial risks and make informed decisions about them.

Problem Statement

Though existing handles expense tracking and budgeting, the information is not connected in a way that the user understands their financial situation and the steps they should do next. Wealthy Minds solves this with organized financial data, identification of critical financial issues, and personalized recommendations.

Literature Review

YNAB supports spending and savings targets and includes a loan planner for testing repayment scenarios (YNAB, 2025, 2026). Rocket Money combines budgeting, spending and subscription monitoring, while selected advanced facilities are offered through Premium (Rocket Money, 2026). Empower presents account-level cash flow, budgeting and portfolio information (Empower, 2026). Although these applications are good at tracking finances, Wealthy Minds takes a step forward by analyzing spending behavior, providing future financial predictions, and using an efficient data structure to deliver meaningful insights.

Table 1: Existing Tools and Capabilities Gap

Existing tool	Established capability	Gap addressed by Wealthy Minds
YNAB	Targets and loan scenarios	Unified behavioural score and explainable prediction
Rocket Money	Budget and subscription monitoring	Transparent prioritisation logic and education
Empower	Cash flow and portfolio dashboard	Data-structure-driven relationship analysis

Proposed Solution

In this system, accounts, income, loans and saving goals represent vertices in a directed weighted graph. The movement of money between the is shown using connections, where each connection stores how much money is transferred. The graph is stored using adjacency lists, while BFS and DFS analyzes the financial connections and track the cash flow. This shows the user exactly how their money moves between financial activities.

System Functionalities

- Stores income, expenses, savings, investments, loans and financial goals.
- Produces monthly, quarterly, and yearly financial reports.
- Calculates financial health score an explains the results.
- Predicts expenses, gives alerts, and suggests recommendations

Table 2: System Flow - Input, Process, Output

Input	Processing	Output
Transaction, amount, category, date	Validation, storage and selected-structure operation	Searchable financial record
Accounts, loans and goals	Behaviour analysis and health scoring	Score, trend and explanation
Historical spending	Forecasting and threshold checks	Prediction, alert and recommendation

Selected Data Structure and Justification

A graph is the most suitable for many to many financial relationships which cannot be represented naturally by linear list or hierarchy. It stores using an adjacency list, while BFS and DFS analyzes

the financial connections and cash flow. Dijkstra's algorithm is used here to find the most efficient financial paths when it's necessary.

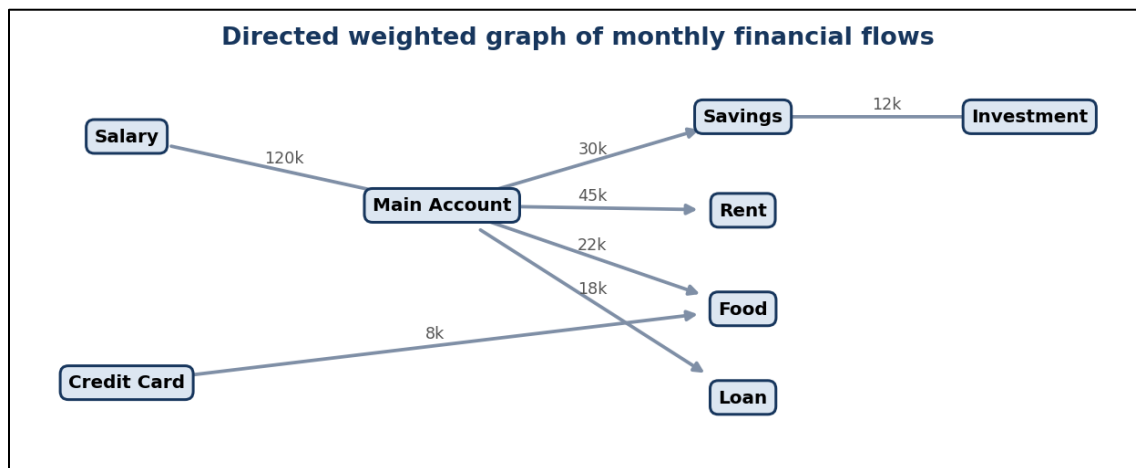


Figure 1: Directed Weighted Graph of monthly finances

Table 3: Operations, Expected and Worst Case

Operation	Expected / typical	Worst case
Add vertex	$O(1)$	$O(1)$
Add edge	$O(1)$	$O(1)$
BFS / DFS	$O(V + E)$	$O(V + E)$
Dijkstra path	$O((V+E) \log V)$	$O((V+E) \log V)$

Novel Features

- Income Distribution Tracker
- Debt and Subscription Monitor
- Savings Goal Planner

Technologies Used

The system is developed using Java. Financial data will be managed by classes like FinancialNode, FlowEdge, FinanceGraph, and FlowAnalyzer with a HashMap used to store the graph connections. JavaFX will display the financial graph and Junit will be used for testing. JGraphT will support the graph algorithms like BFS, DFS, and Dijkstra's.

Selling Points of the Application

- Providing personalized financial advice instead of just recording transactions
- Explains how the financial health score and alerts are produced.
- Designed specifically for young adults and students.
- Using graph data structure for faster and insightful financial analysis.

Individual Reflection

My contribution to this project was implementing the financial data using a graph, the adjacency list, and developing the BFS and DFS to analyze how the money flows. With that, I learned how graphs can represent real world financial relationships. I also gained a better understanding of graph traversal and how it is used to analyze financial data and produce meaningful insights. During development, one of the main challenge was deciding how different financial entities should be connected so that the graph can accurately represent the movement of money. I also had to ensure that the graph handled duplicate connections that could affect the analysis.

Repository Links

- **GitHub Profile:** <https://github.com/Nimthaka808>
- **GitHub Repository:** <https://github.com/Daham728/WealthyMinds>

Conclusion

The module shows how a data structure can be used to solve real world problem. In Wealthy Minds, it helps analyze financial data, recognize connections between transactions and make informed decisions. Because of functionalities like financial predictions, health scoring, personalized recommendations, the system offers more than a simple expense tracker.

Implementation Validation Plan

- Test the system with valid, invalid, null, duplicate data.
- Check results using a sample dataset with known outputs.
- Measure how system performance when load increases.
- Request user feedback upon testing the system.

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